MILITARY SPECIFICATION

SPIROMETER, PORTABLE, 5.0 LITER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The recording spirometer covered by this specification shall be a diagnostic instrument used to determine ventilatory function and permit the measurement of lung volumes and of time-volume relationships.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	- Box, Folding Paperboard.
PPP-B-585	- Box, Wood, Wirebound.
PPP-B-601	- Box, Wood, Cleated-Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Fiberboard.
PPP-B-676	- Box, Set-Up, Paperboard.

MILITARY

MIL-M-7298	- Manual, Technical, Commercial Equipment.
MIL-L-10547	- Liners, Case, And Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, Code ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6515

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

No. 25020 Spirometer, Portable with Chart and Mouthpiece

(Copies of specifications, standards, drawings, and publications required in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Underwriters Laboratories Inc. (UL) Standards for Safety

UL 544 Medical and Dental Equipment, Electric

(Application for copies should be addressed to the Underwriters Laboratories, Inc., Publications Stock, 333 Pfingsten Road, Northbrook, IL 60062.)

3. REQUIREMENTS

3.1 Materials. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the product. When a definite material is not specified, a material shall be used which will enable the instrument to meet the requirements of this specification.

3.1.1 Metal parts. All metal parts shall be of a corrosion resistant material or treated in a manner to render them adequately resistant to corrosion.

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3.1.2 Other materials. Materials shall conform to applicable specifications and shall be as specified herein. Materials for which there are no applicable specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight and suitable for the purpose intended.

3.1.3 Elastomeric material. The bellows body shall be fabricated from rubber or other suitable non-porous and flexible sheet material.

3.1.4 Dissimilar metals. Dissimilar metals shall not be placed in intimate contact with each other unless suitably protected against electrolytic corrosion with protective coatings.

3.1.5 Protective treatment. When materials are used in the construction of the instrument that are subject to atmospheric or environmental conditions likely to cause corrosion in normal service life, they shall be protected against corrosion that will in no way prevent compliance with the performance requirements of this specification. Finishes and protective coatings which will crack, chip, or scale during normal service life or are affected by extremes of atmospheric or environmental conditions, shall not be used.

3.1.6 Fumes and vapors. Materials used in the construction of the instrument shall not produce corrosive, deleterious, or toxic fumes or vapors under the conditions specified herein.

3.1.7 Fungus-proof materials. Materials which are nutrients for fungi shall not be used where it is practicable to avoid them.

3.2 Design and description. The instrument shall be a portable respiration testing device which records a spirogram over a specified time interval and provides a permanent graphic record, from which the forced vital capacity (FVC) and forced expiratory flow (FEF) can be calculated. It shall be a device having an outer housing, a bellows within the housing, a breathing tube extension of the bellows, and a recording mechanism attached to the case which shall sense and record the expansion and contraction of the bellows. The instrument shall be so constructed that no parts will work loose in service and so that it will withstand the normal shocks, vibrations, and such other conditions as are incident to service, shipping, storage, and installation.

3.2.1 Physical dimensions. Maximum envelope dimensions of the instrument shall not exceed those specified in Figure 1.

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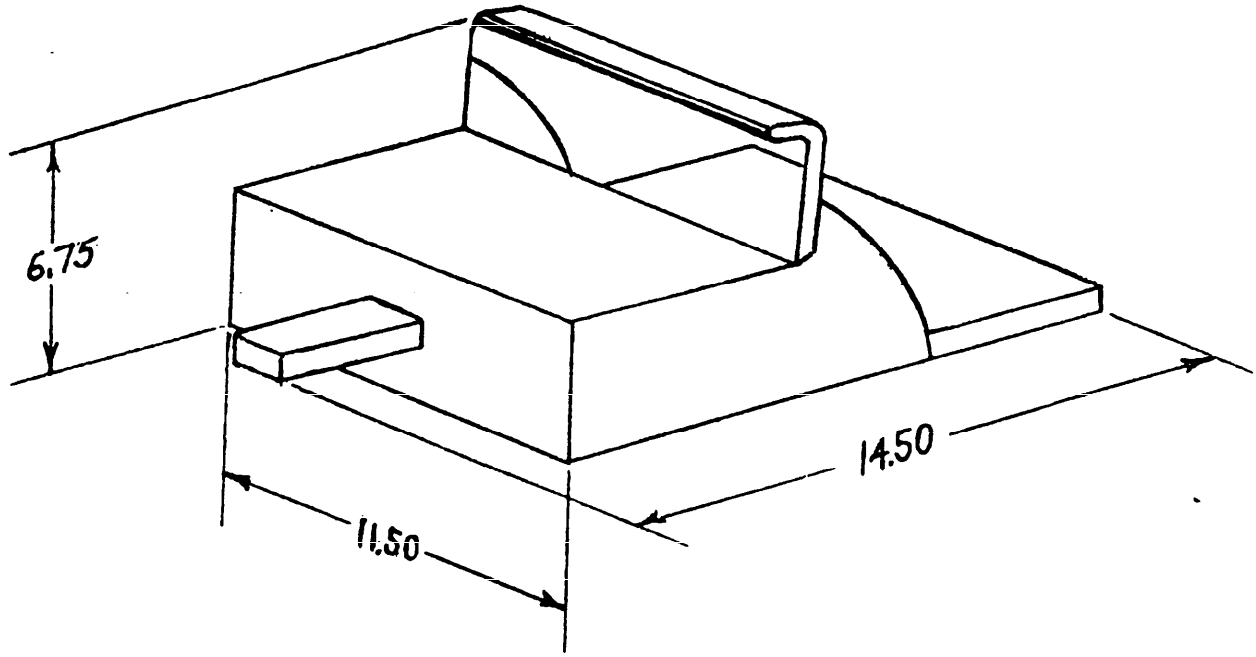


FIGURE 1. **Maximum envelope dimensions**

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3.2.2 Housing. The housing shall be designed to provide a suitable enclosure, mounting surfaces and attachment points for the functional component parts of the instrument. It shall be made of nonferrous metal or plastic, or a combination of both, uniform in texture and having a smooth surface. When required, the housing shall be finished with a durable type material in accordance with the requirement of 3.1.5 and which will withstand usage encountered in service.

3.2.3 Chart drive and timer assembly. The chart drive and timer assembly shall be securely mounted in such a manner that it will not loosen, slip or become misaligned during operation. The method of attachment shall not interfere with and shall allow proper alignment of the chart carriage spur rack and motor spur gear.

3.2.3.1 Chart drive motor. The chart drive motor shall be rated at 115-V/60 Hz and shall be designed to drive the chart carrier in a horizontal plane at the constant linear rate specified in 3.3.2.

3.2.3.2 Motor spur gear. The motor spur gear shall be designed to ensure full face contact with the mating carriage spur rack without requiring critical adjustment. Backlash shall be held to a minimum consistent with the accuracy of the recording mechanism.

3.2.4 Chart drive activation system. The chart drive activation system shall consist of a pushbutton switch and an air operated sensor which shall control the synchronous drive motor. It shall be designed to minimize the time lag between application of air pressure to the bellows (beginning of expiration) and the timer drives synchronous timing.

3.2.4.1 Ready-reset switch. A pushbutton switch shall be located on the right side of the instrument case. It shall be normally open and shall control the chart drive motor in conjunction with the air operated sensor. A positive indication of control activation shall be provided (e.g., snap feel, audible click, or integral light).

3.2.4.2 Air operated sensor. The air operated sensor shall automatically activate the drive motor any time after the ready-reset switch is actuated (depressed). It shall be designed so that the drive motor does not start until air is entering the mouthpiece of the instrument and a pressure differential is detected.

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3.2.5 Chart holder and carrier assembly. The chart holder shall contain the spirometer pressure sensitive chart and shall have a carriage spur rack that will mesh with the drive motor spur gear. Chart retainer clips shall be integral with the chart holder and shall be designed to facilitate chart insertion and removal to and from the chart holder carrier assembly. The chart holder shall be designed to provide positive chart alignment using two locater pins or an equivalent method. The design of the carriage assembly and guide rails shall provide for alignment of the chart holder carrier and the stylus assembly at time and volume zero.

3.2.6 Bellows and breathing tube assembly.

3.2.6.1 Bellows. The bellows shall have a minimum capacity of 5 liters and be a pleated type suitably designed for breathing diagnostic tests. The accordion pleated body shall be suitable for "cold" sterilization by commonly used antibacterial detergents without requiring removal of the bellows assembly from the unit. Alternatively, the bellows and breathing tube assembly shall be removeable from the unit for sterilization purposes, and shall be marked "TOP" on the top surface to facilitate disassembly and reassembly. The bellows shall be designed so that expansion and contraction takes place in the vertical (y) plane for a pivotal movement about a fixed point, which shall be aligned with the stylus assembly. Alternatively, the bellows may be mounted in the horizontal (x) plane, provided it is so designed to minimize or eliminate the inherent time lag required for the bellows to expand under pressure. The time delay being defined as from the time (T_o) air enters the bellows, to the time (T_n), the pressure of air entering the bellows overcomes the weight of the bellows and the inertia of the stylus mechanical linkage and supporting bellows end plate. Bellows dead space prior to expansion shall not exceed 200 cc of air, when tested as specified in 4.5.1.3. Under pressure, the bellows shall not leak and there shall be no quivering or bouncing of the bellows.

3.2.6.2 Breathing tube. The breathing tube shall be designed to extend beyond the front panel of the unit and shall be connected to the internal bellows volume. It shall be designed so that the cross-sectional area does not restrict air flow. The portion of the tube extending beyond the front panel shall accept the disposable mouthpiece.

3.2.6.3 Breathing tube mouthpiece. The breathing tube mouthpiece shall be of a disposable type and shall fit onto the external portion of the breathing tube. It shall be fabricated from cardboard or other suitable material. Style, design and dimensions shall be in accordance with DPSC Drawing No. 25020. Any alternative design shall conform to sound spirometer design practice. The cross-sectional area of the mouthpiece shall be compatible with that of the breathing tube and shall not impede, restrict or increase resistance to airflow. Twenty-five (25) disposable mouthpieces shall be furnished with each instrument.

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3.2.7 Stylus assembly. The stylus shall scribe a readable impression on the spirometer pressure sensitive chart over the entire volume and time range specified. It shall be mechanically attached to the stylus arm holder and bellows end plate or alignment shaft. The assembly shall be aligned to and rotate about or move horizontally with the bellows alignment shaft under bellows action. The radius of arc through which the end plate moves (distance between pivot point and end plate) or the horizontal movement of the end plate shall be selected to provide the degree of accuracy necessary to meet the requirements of this specification.

3.2.8 Spirometer chart. The spirometer chart shall be a hard coat, high contrast material of the pressure sensitive type, suitable for recording and providing a permanent record of the pulmonary functions specified in 3.2. The chart design shall be such that the flow rate scale shall provide direct readings of values anywhere along the flow curve of the time-volume relationships. The time axis shall be in the horizontal plane and the amp itude (volume) axis shall be in the vertical. The input span shall be 0 to 5 liters minimum over a time interval of 6 seconds. The time and amplitude axis shall be in accordance with the requirements of Table 1.

TABLE I. Chart design requirements

Time Axis		Amplitude Axis	
Major Division every:	20 mm	Major Division:	5
Minor Div./Major Div:	10	Minor Division:	10
Grid Width:	120 mm	Numbers:	0 to 5

The chart shall be compensated for bellows dead space, if required. The chart shall be capable of being mounted on the chart holder and carrier assembly. Style, design and dimensions may be in accordance with DPSC Drawing 25020. Each instrument shall be furnished with 100 pressure sensitive charts.

3.2.9 Electrical power supply system.

3.2.9.1 General. Switches, attachment plugs, receptacles and other components handled by the user shall be mounted securely. Terminals solely for connection of equipment-grounding conductors shall be in accordance with UL 544. Wiring terminals shall be provided with appropriate soldering lugs or pressure terminals securely fastened in place and shall be prevented from turning. Minimum conductor wire size shall be No. 18 AWG for branch circuits. Power cords shall be as specified in 3.2.9.2. Strain relief shall be provided on the power cord so that the cord shall be capable of withstanding, for one minute without displacement, a pull of 35 pounds, with the connections within the instrument disconnected. Other than terminal rods and terminal plates, unplated iron or steel shall not be used for the current-carrying parts of the device. Internal wiring shall be routed away from sharp edges,

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screw threads and other abrading conditions. Switches and other control devices shall have ratings not less than that of the lead controlled. The current-carrying capacity of a switch which controls an inductive load shall be not less than twice the rated full-load current of the inductive device.

3.2.9.2 Power supply cord. The power supply cord shall be 18 gauge, three-wire conductor terminating in a 15A, 115-V, two-pole, three-wire, hospital grade grounding plug, conforming to applicable UL Standards. The cord shall have a minimum length of 8 feet.

3.2.9.3 Power supply on-off switch. The instrument may be designed to incorporate a power supply "on-off" switch. It shall be a toggle switch or other suitable device with two discrete positions. The switch shall be vertically oriented on the front panel of the instrument, with "off" in the down position. A simple indicator light, such as a pilot, bull's-eye or jewel shall be an integral part of or located adjacent to the switch. Green shall indicate "power-on".

3.2.9.4 Other electrical components. All other electrical components shall be suitable for the use intended. They shall conform to the requirements of the applicable UL Standard, or be UL-listed, where a standard or approval is available.

3.2.10 Vital capacity table. A table of accepted vital capacity standards (norms) for men and women shall be permanently attached to the instrument.

3.2.11 Instrument carrying case. The instrument shall be supplied with a carrying case fabricated from corrugated fiberboard or other suitable material. It shall have a suitable carrying handle and shall be designed to include inserts in the case to accommodate all the component parts specified. The inserts shall prevent movement during transit, shifting and damage during carrying and when the set is not in use.

3.3 Performance

3.3.1 Functional operation. When tested as specified in 4.5.1.1, the instrument shall function as intended and meet the requirements specified herein.

3.3.2 Chart carrier linear rate. The chart carrier and carriage assembly shall be driven by the chart drive motor at a rate of 20 mm $\pm$ 2 mm per second. The time required to travel 120 mm shall not exceed 6 $\pm$ 0.1 seconds when tested as specified in 4.5.1.1.

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3.3.3 Stylus trace and repeatability. When tested as specified in 4.5.1.2 the stylus shall strike the pressure sensitive chart and scribe a readable, clearly defined impression transversely across the entire grid width. The trace scribed by the stylus shall be repeatable within ± 10 %.

3.3.4 Stylus response. When tested as specified in 4.5.1.3, the volume of air required to make the stylus begin to rise shall not exceed 200 cc. Rise time shall not exceed 0.3 seconds.

3.3.5 Stylus deflection. Full scale deflection of the stylus shall occur within 5 seconds when tested in accordance with 4.5.1.4.

3.3.6 Mechanical noise and trace fluctuations. When tested as specified in 4.5.1.1, the carrier assembly shall operate smoothly. There shall be no excessive mechanical noise or gear chatter. The bellows shall not fluctuate under pressure and cause discontinuities of the trace.

3.4 Service data. The contractor shall furnish, with each unit, two copies each of a technical manual which contains complete instructions for installation, operation, maintenance, and lists of component parts. The manual may comply with the requirements of MIL-M-7298. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via cognizant Government inspector, two preliminary copies of his proposed technical manual for review and comment by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of any required revisions and corrections in the final manual without delaying delivery under the contract, or purchase order. In addition, the contractor shall furnish 15 copies of the approved technical manual to the contracting officer for distribution. Should be supplier have furnished acceptable manuals for the identical item within two years preceding the date of the contract and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request to the contracting officer. Waiver of submission of drafts and submission of 15 additional copies shall be at the option of the Government.

3.5 Identification marking. A metal nameplate or other suitable permanent type identification marking, as specified below, shall be fastened to each instrument. The pressure sensitive chart and disposable mouthpiece shall also be marked with the manufacturer's name, address and catalog number as well as the National Stock Number (NSN). The location of the marking on the mouthpiece shall be selected so that it does not come in contact with the patient's mouth.

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3.5.2 Manufacturer's identification. The nameplate shall be legible and durably imprinted with the following: model number, serial number, voltage, phase, cycles and maximum power. It shall be affixed to the outside of the instrument front panel or other accessible.

3.6 Interchangeability and substitutability. Interchangeable parts having the same manufacturer's part number shall be functionally and physically interchangeable. Component parts except for the recording chart need not be interchangeable between manufacturers and may not be interchangeable with different models produced by the same manufacturer.

3.7 Weight. The weight of the complete instrument shall not be greater than 5.75 pounds.

3.8 Fire and casualty hazards. The spirometer shall comply with the fire and casualty hazards provisions of the applicable standards and be an approved type for the purpose intended, as determined by a nationally recognized testing agency, adequately equipped and competent to perform such services, and as shall be evidenced by the attachment of its seal, stamp or label to the article, or issuance of a certificate. The stamp, label, seal or certificate of the Underwriters' Laboratories (New York, NY), the United States Testing Laboratories, Inc. (Hoboken, NJ), the Electrical Testing Laboratories, Inc. (New York, NY), or other testing laboratory approved by the contracting agency shall be considered acceptable evidence of such approval; however, in addition to the aforementioned, the contracting agency reserves the right to require the contractor to submit a laboratory test report which shall include test results and methods used to determine conformance.

3.9 Recording mechanism adjustment. The pressure the stylus exerts on contact with the chart surface shall be capable of being varied to meet the requirement specified in 3.2.5. A means shall be provided for zero control (alignment of the stylus and chart origin) in both the "x" and "y" direction. It shall be such that if horizontal movement of the carrier before the stylus begins to rise or vertical movement of the stylus before the carrier begins to move, occurs, the instrument can be adjusted so that the curve begins at the chart origin.

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3.10 Workmanship. Workmanship shall be first class throughout. The instrument, pressure sensitive chart and disposable mouthpiece shall be free from defects which detract from their appearance or impair their serviceability, proper function, and intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the instrument may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the National Stock Number (NSN), the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 Inspection lot. The inspection lot shall be determined in accordance with MIL-STD-105.

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4.2.2 Preparation for delivery. The inspection lot size shall be expressed in units of one fully prepared shipping container, containing complete instruments fully prepared for delivery and made from essentially the same materials and components. The sample unit shall be one fully prepared shipping container, containing instruments fully prepared for delivery, with the exception that it need not be sealed. The inspection shall be Inspection Level S-2 and the Acceptable Quality Level shall be 4.0 defects per hundred units in accordance with MIL-STD-105.

4.2.3 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and as indicated in table II. The unit of product shall be one complete spirometer.

TABLE II. Sampling for examination

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor	II	2.5

4.2.4 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table III. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Unless otherwise specified the unit of product for tests shall be one complete spirometer set.

TABLE III. Sampling criteria for tests

Item	Characteristic	Requirement	Test procedure	Inspection level	AQL
Spirometer	Functional operation	3.3.1	4.5.1.1	S-2	1.0
	Chart carrier linear rate	3.3.2	4.5.1.1		
	Stylus trace and repeatability	3.2.6.1 3.3.3	4.5.1.2		
	Stylus response	3.2.6.1 3.3.4	4.5.1.3		
	Stylus deflection	3.2.6.1 3.3.5	4.5.1.4		
	Mechanical noise and trace fluctuation	3.2.6.1 3.3.6	4.5.1.1		
Breathing tube & bellows assembly	Bellows leakage	3.2.6.1	4.5.2		

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4.3 Examination. The instrument shall be examined to determine compliance with all requirements contained in this specification. Examination shall be conducted as specified in tables II and IV. Examination, however, shall not be restricted to defects listed in table IV. Any spirometers in the sample containing one or more defects shall be rejected and if the number of defective spirometers in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

TABLE IV. Classification of defects

Categories	defects
Critical	None defined.
Major	
101	Disposable mouthpieces missing or quantity not as specified.
102	Pressure sensitive charts missing or quantity not as specified.
103	Breathing tube not securely attached to internal bellows volume.
104	Instrument physical dimensions exceed maximum allowable envelope.
105	Disposable mouthpiece physical dimensions not as specified.
106	Disposable mouthpiece does not fit on breathing tube.
107	Pressure sensitive chart time and amplitude axis not as specified.
108	Instrument has exposed sharp edges.
109	Power cord missing, defective, or not as specified.
110	Stylus holder deformed or misaligned.
111	Stylus not securely held in place or adjustable for pressure.
112	Spur rack and spur gear do not mesh properly.
113	Spur rack and spur gear teeth not fully cut or developed.
114	Total weight exceeds maximum allowable.
115	Installation, operation and maintenance data not furnished.
116	Underwriter's Laboratory certification or equivalent missing.
117	Bellows not removeable.
118	Carrying case not furnished.
119	Vital capacity charts not permanently attached to unit.
120	Electrical components and wiring not shielded or enclosed in case.
121	Chart retainers inoperative or missing.
122	Identification marking of instrument, chart and mouthpiece not as specified.

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TABLE IV. Classification of defects (continued)

Categories	Defects
MINOR	
201	Identification marking of carrying case not as specified.
202	Bellows marking "TOP" missing.

4.4 Examination for preparation for delivery. Each of the fully prepared shipping containers, containing instruments, selected as a sample unit from the lot shall be visually examined to determine that the packaging, packing and marking conform to this specification. The classification of defects, Table V, shall be used to classify the defects found.

TABLE V. Classification of defects for preparation for delivery

Examine	Defects
Markings	Omitted, incorrect, illegible, or improper size, location, sequence or method of application.
Packaging and packing materials	Any component missing; any component damaged, affecting serviceability.
Workmanship	Inadequate application of components such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Weight of content exterior and interior	Number of intermediate packages is more or less than required; gross/net weight exceed requirements.

4.5 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.5.1 Performance tests. All performance tests shall be made with the instrument connected to a 115-V, 60 Hz power source. A regulated air supply under pressure, and air release valve shall be connected directly to the external portion of the breathing tube connected to the internal bellows volume. Appropriate instrumentation to indicate parameters specified and data required shall be included in the test set-up.

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4.5.1.1 Chart carrier linear rate and operational checkout. With power on and the carrier at the "zero" position, depress the ready-reset push-button. There shall be no carrier or stylus movement. Open the regulated air supply connected to the bellows and supply a sufficient volune of air to overcome bellows dead space (do not exceed 5.0 liter capacity of bellows). With the ready-reset switch still depressed the carrier shall travel its full excursion and the stylus shall rise as the bellows expands. With an appropriate timing device, measure chart carrier linear rate. Release ready-reset switch and return carrier to "zero" position. With air supplied to bellows depress ready-reset switch for 2 seconds and release. Carrier shall cease to move. With air still supplied to bellows, return carrier to "zero" position, depress ready-reset switch and bleed air from bellows in less than 5 seconds. Carrier shall stop movement. During performance of the operational checkout all components shall perform as intended and there shall be no excessive mechanical noise or gear chatter.

4.5.1.2 Stylus trace and repeatability. Insert a pressure sensitive spirometer chart on the chart carrier, adjust stylus pressure and return carrier to "zero" position so that stylus and chart origin are aligned. With power on, depress the ready-reset switch and supply a total volume of 3 liters of air to the bellows at the rate of 0.5 liters per second for 6 seconds. The stylus shall scribe a readable impression transversely across the entire grid width of the chart. Return carrier to "zero" position and repeat procedure three times. The trace scribed by the stylus shall be repeatable within $\pm$ 10%. There shall be no discernable trace " glitches" or discontinuities.

4.5.1.3 Stylus response. Insert a pressure sensitive chart on the chart carrier, adjust stylus pressure and return carrier to "zero" position so that stylus and chart are aligned. With power on, depress the ready-reset switch and supply a sufficient volume of air to the bellows to cause the stylus to begin to just rise. The volume of air required to make the stylus rise (bellows dead space) shall not exceed 200 cc, and shall occur in no more than 0.3 seconds.

4.5.1.4 Stylus deflection. Insert a pressure sensitive chart on the chart carrier, adjust stylus pressure and return carrier to "zero" position so that stylus and chart origin are aligned. With power on, depress ready-reset switch, and supply a total volume of 5 liters of air to the bellows at the rate of 1.0 liter per second for 5.0 seconds. Trace scribed shall be a straight line indicating full scale stylus deflection of 5.0 liters.

4.5.2 Bellows leakage. A regulated air supply under pressure shall be connected directly to the external portion of the breathing tube connected to the internal bellows volume. Supply the bellows with a total volume of 4.0 liters of air and hold for 5 minutes. There shall be no indication of bellows leakage. This test shall be performed on the breathing tube and bellows assembly, or at the option of the manufacturer may be performed on the complete end item.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.1).

5.1.1 Unit of issue. One spirometer with carrying case, 100 charts and 25 mouthpieces, or one package containing 100 charts or one package containing 25 mouthpieces as specified, constitutes one unit of issue.

5.1.1.1 Level A.

5.1.1.1.1 Unit package.

5.1.1.1.1.1 Spirometer. Each spirometer shall be supplied in a carrying case in accordance with 3.2.11. Each complete unit shall be further packaged in a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening.

5.1.1.1.1.2 Charts and mouthpieces. One unit of issue of charts or one unit of issue of mouthpieces shall be packaged in a box of appropriate size and constructed in accordance with PPP-B-566, PPP-B-676 or PPP-B-636, type CF, class domestic. Closure of box shall be adequate to prevent accidental opening.

5.1.1.1.2 Intermediate package. Twelve units of issue of charts shall be packaged in a box of appropriate size and constructed in accordance with PPP-B-636, type CF, class domestic. Closure of box shall be adequate to prevent accidental opening. Intermediate package is not required for spirometers or mouthpieces.

5.1.1.2 Level C. Units of issue shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. Six (6) units of issue of spirometers or seventy two (72) units of issue of charts or twenty (20) units of issue of mouthpieces shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather resistant, grade V3c. Closure and strapping shall be as specified in the appendix of the applicable box specification.

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5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof as specified in the appendix of PPP-B-636.

5.2.2 Level B. Six (6) units of issue of spirometers or seventy two (72) units of issue of charts or twenty (20) units of issue of mouthpieces shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type: PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. Units of issue shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Unit and intermediate package and exterior container. Each unit and intermediate package and each exterior container shall be marked in accordance with MIL-STD-129. The legend "HANDLE WITH CARE" shall be marked on the unit and exterior containers of the spirometers.,

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options specified herein, and the procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Applicable levels of packaging and packing, (see 5.1 and 5.2).
(c) National Stock Number

6.2 National Stock Number. This specification is intended to cover the following item appearing in the Federal Supply Catalog:

National Stock No.	Item Identification
6515-00- 089-5025	SPIROMETER, PORTABLE, 5.0 Liter, 115 Volt, 60 Hz, AC
6515-00-089-5026	CHART, SPIROMETER
6515-00-089-5027	MOUTHPIECE, SPIROMETER, Cardboard, Disposable

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6.3 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of this specification, or which are commercially availble, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.4 Definitions. Words, terms and expressions used in this specification which are peculiar to the general field of ventilatory function testing are presented in the table below and represents accepted terms for certain ventilatory measurements.

TABLE VI. Ventilatory function tests, symbol and definition

Description	Term used	Symbol	Remarks
The largest volume measured on complete expiration after the deepest inspiration without forced or rapid effort.	Vital Capacity.	VC	
The vital capacity performed with expiration as forceful and rapid as possible	Forced vital capacity.	FVC	
Volume of gas exhaled over a given time interval during the performance of a forced vital capacity.	Forced expiratory volume (qualified by subscript indicating the time interval in seconds).	FEV_T ($FEV_{1.0}$)	
FEV_T expressed as a percentage of the forced vital capacity: $\frac{FEV_T}{FVC} \times 100$	Percentage expired (in T seconds).	$FEV_{T\%}$	Time-volume relationship
The average rate of flow for a specified portion of the forced expiratory volume, usually between 200 and 1200 ml.	Forced expiratory flow.	$FEF_{200-1200}$	Formerly called maximum expiratory flow rate (MEFR).

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TABLE VI. Ventilatory function tests, symbol and definition (Continued)

Description	Term used	Symbol	Remarks
Average rate of flow during the middle half of the forced expiratory volume	Forced mid-expiratory flow.	$FEF_{25-75\%}$	Formerly called maximum mid-expiratory flow rate.
Volume of air which a subject can breathe with voluntary maximal effort for a given time.	Maximal voluntary ventilation.	MVV	Formerly called maximum breathing capacity.

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Preparing activity:
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DOCUMENT IDENTIFIER AND TITLE

MIL-S-37227 SPIROMETER, PORTABLE, 5.0 LITER

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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